Week 11a: Custom Secure Boot + AES in Firmware: Building a CAC-Signed Bootloader and Reverse Engineering It

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What You'll Learn This Week

By the end of this tutorial, you will be able to:

- Understand why consumer and industrial devices use a secure (authenticated) boot chain
- Explain, with a diagram, how AES-128-CMAC (RFC 4493) authenticates a firmware image
- Read a custom linker script, modify the MEMORY block, and relocate a program to a higher flash address
- Inspect a relocated ELF with `readelf`, `nm`, and `objdump` to confirm where every section and symbol lives
- Use Mbed TLS (already shipped inside the Pico SDK) in your own C code to decrypt data at runtime
- Flash a raw `.bin`, a manifest, and an application to three different flash addresses with OpenOCD
- Tamper with a single byte of firmware and watch the bootloader reject it at boot
- Reverse engineer the bootloader with GDB live on the Pico 2 (breakpoints, memory dumps, single-stepping the jump)
- Reverse engineer both firmware images in Ghidra, including recovering an AES key, IV, and ciphertext from flash
- Explain why a static key is not a real secret, and how real products solve secure boot (OTP keys, ROM verification, ECDSA)

Part 1: Why Secure Boot? (The Threat Model)

Firmware is a Product, and Products Get Modified

Everything you have flashed onto your Pico 2 so far in this course has been a **single standalone image**. The RP2350 boot ROM reads the image at `0x10000000`, your program runs, done. That is exactly how millions of cheap embedded products work, and it has a serious weakness:

Anybody with a $5 probe can replace the flash contents.

OpenOCD, a 5-pin SWD connection, and a serial terminal are all an attacker needs to:

1. Dump your firmware (read flash)
2. Modify a byte (change a password, unlock a feature, bypass a license)
3. Flash their own image (rootkit the device)

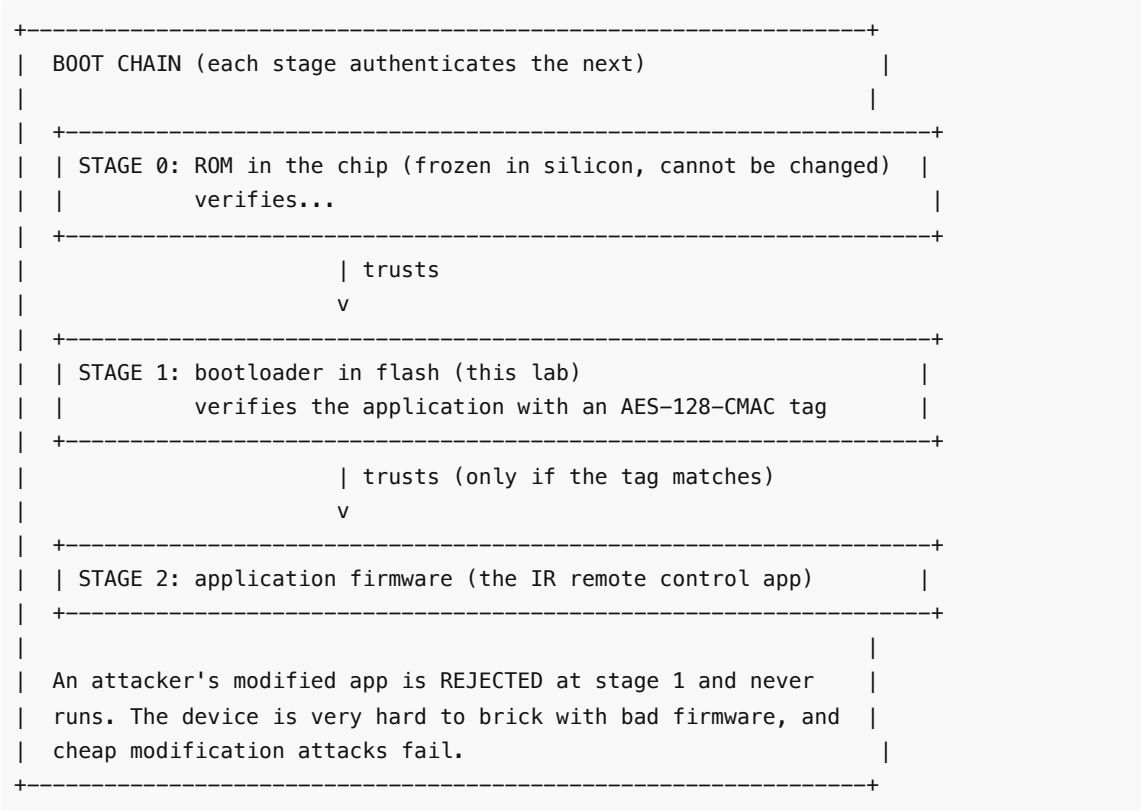
Now tighten the screw. Imagine your product is:

- A smart lock that talks to the network
- A security camera that signs video
- A commercial battery charger that detects counterfeit batteries
- An update server that pushes "fixes" to thousands of deployed devices

In those products, **running modified or unauthorized firmware is a real-world safety and business problem**. If a device can be made to run code it did not originate, that device cannot be trusted, and neither can the data it produces.

The Defense: A Verified Boot Chain

The modern defense is a **boot chain**. Each stage of the chain refuses to start the next stage until it has proven (cryptographically) that the next stage is authentic:



> ***What this lab does, and what it does not.***
>
> *The 0x0026a bootloader authenticates the application using a key that is **embedded in the bootloader itself**. That stops a casual attacker who uses OpenOCD to patch a byte or two. A serious attacker who dumps the flash can read the key and re-sign their own image. Real products push Stage 0 into the ROM*

> *and store the key in **OTP (one-time programmable) memory** that flash reads cannot access. Part 19 shows how to think about that gap.*

Part 2: Choosing the Authentication Primitive

The bootloader must answer one question about the application image: "Did this byte-for-byte image come from the person who holds the key?"

The Candidate Tools

Tool	What it computes	Can an attacker forge it?
CRC-32	A simple checksum	YES, trivially. CRC is designed to catch noise, not attackers.
SHA-256 hash	A one-way digest of the image	YES, if there is no secret. Anyone can compute the hash after editing.
HMAC-SHA256	A keyed digest	NO (without the key). Strong, but needs a keyed hash implementation.
AES-CMAC	A keyed MAC built from a block cipher	NO (without the key). Uses the AES primitive you already have.
ECDSA signature	An asymmetric signature	NO. The verifier holds only a public key, so even a full flash dump cannot forge.

Why AES-CMAC for This Lab

Two practical reasons:

1. **The Pico SDK already ships a fast, well-tested AES.** CMAC is built on AES (it is a CBC-MAC variant), so we do not add SHA code or HMAC glue; we reuse the `mbedtls` library that ships inside the SDK (`pico_mbedtls`).
2. **One primitive, two lessons.** The same AES engine that authenticates the image in the bootloader is also used inside the application to decrypt a secret banner (Part 9). You see one cipher doing two classic jobs.

ECDSA would be stronger (asymmetric, so the flash dump does not reveal the signing key), and the RP2350 ROM ships a `secp256k1` verifier. The gap between where this lab leaves off and where the ROM verifier lies is an important "real product" paragraph in Part 19.

Part 3: AES-128-CMAC in Depth (RFC 4493)

A MAC Is a Keyed Checksum

A **Message Authentication Code (MAC)** takes three inputs:

- a secret key `K`
- a message `M`
- (optionally) a length `Tlen`

and produces a short, fixed-size **tag** `T` . Security property: without `K` , an attacker cannot construct `(M, T)` that verifies, even if they are allowed to see thousands of valid `(M, T)` pairs.

Outline of CMAC

CMAC is a keyed block-cipher MAC standardised in **NIST SP 800-38B** and implemented in **RFC 4493** for AES-128. It is a **CBC-MAC with a twist**: the final block is XORed with one of two derived subkeys before the last AES encryption, which removes the classical CBC-MAC extension attack.

```
+-------------------------------------------------------------+
|  Subkey derivation (once, at key setup)                     |
|                                                             |
|  L = AES_K(0^128)                    AES-encrypt a zero block  |
|  if MSB(L) == 0: K1 = L << 1                                |
|  else:          K1 = (L << 1) XOR 0x87    (0x87 = Rb for AES)   |
|  same step on K1 yields K2                                  |
|                                                             |
|  Tagging (per message M, split into 128-bit blocks M1..Mn)      |
|                                                             |
|  M0 = 0^128                                                 |
|  if M is an exact multiple of 128 bits:                     |
|        Mn' = Mn  XOR K1                                     |
|  else:  (pad M with 10...0 to a full block)                 |
|        Mn' = padded(Mn) XOR K2                              |
|                                                             |
|  for each block:  C[i] = AES_K( C[i-1] XOR M[i] )             |
|  Tag = C[n] (first Tlen bits)                               |
+-------------------------------------------------------------+
```

The two subkeys exist precisely to encode "full last block" vs "short last block" into the tag. If we reused plain CBC-MAC, an attacker could cut the message after a block boundary and still produce a valid tag.

Where Is AES-128-CMAC Not Really "Analysis" But Simple Math?

Study the equation for a single message block:

```
C[1] = AES_K( 0^128 XOR M1 )
```

Because this is a block cipher, a 1-bit change anywhere in `M1` results in a completely unrelated tag. And without `K`, you cannot compute `AES_K(...)` at all. Both facts are what make the tag unforgeable in practice.

The RFC 4493 Test Vectors (how the tool proves itself)

The host signer `tools/sign_firmware.py` implements AES and CMAC from scratch in pure Python, so the build toolchain has zero extra dependency on the development machine. It is validated against the four official test vectors from RFC 4493 using the well-known test key:

```
Key  = 2b7e151628aed2a6abf7158809cf4f3c

Message = (empty)
Tag     = bb1d6929e95937287fa37d129b756746

Message = 6bc1bee22e409f96e93d7e117393172a
Tag     = 070a16b46b4d4144f79bdd9dd04a287c
```

```
Message = 40-byte repeating sample
Tag     = dfa66747de9ae63030ca32611497c827

Message = 64-byte repeating sample
Tag     = 51f0bebf7e3b9d92fc49741779363cfe
```

Read the tool's top-of-file comment; the four vectors are exercised against the signer's internal CMAC routine before any real image is signed.

Part 4: The Hardware and the Flash Map

The Pieces

Component	Role in this lab
Pico 2 (RP2350)	The target. Two Cortex-M33 cores, 512 KB SRAM, executes code in place from XIP flash.
Debug Probe	CMSIS-DAP SWD adapter for flashing, debugging, and UART serial. It owns the SWD port and usually also gives a UART console channel.
IR receiver	GPIO5 input; decodes NEC remote commands in the application.
3 LEDs	GPIO16, GPIO17, GPIO18; driven by decoded commands.
USB-UART/cable	If the Debug Probe's UART is wired to the Pico's UART0, the bootloader and app log to it.

Why 0x10000000?

On the RP2350 the first 4 MiB of the address map (from `0x10000000`) is a **memory-mapped view of external XIP flash** (eXecute In Place). The CPU fetches instructions straight from that window; the flash controller reads the bytes transparently through the XIP cache. The boot ROM boots from this flash window because it is simply memory.

So "flashing at `0x10000000`" means "making the contents of physical flash visible at the addresses your program expects." A raw `.bin` carries no address information; you must always supply the load address out of band (Part 12).

The 0x0026a Flash Layout

```
Physical XIP flash, as addressable memory
+---------------------------------------------- 0x10000000
| BOOTLOADER image  (0x0026a_bootloader.bin)    |
|   ~28 KB of code + data                       |
|   boot.c, verify.c, manifest.c, aes_cmac.c,   |
|   image.c, main.c                             |
+---------------------------------------------- ~0x10006D00
|   (free padding)                              |
+---------------------------------------------- 0x1001F000
| MANIFEST sector (4 KB, exactly one flash      |
|   sector): magic, version, app address,       |
|   app size, 16-byte AES-128-CMAC tag          |
```

```
+---------------------------------------------- 0x10020000
| APPLICATION image (0x0026a_app.bin)           |
|   the Week 11 IR remote control app,          |
|   now with an AES-128-CBC secret banner,      |
|   linked to run from 0x10020000               |
+---------------------------------------------- 0x10020000 + 0x5498
|   (free padding)                              |
+---------------------------------------------- 0x10200000  (2 MiB)
```

Three things to notice:

1. The manifest sits in its **own flash sector** (`0x1001F000`), so updating it never touches the bootloader or the application.
2. The application is **relocated** above both, starting at `0x10020000` . It is not written at `0x10000000` .
3. The bootloader jump at the end of Part 8 hands control to the app's reset vector, which lives at `0x1002015C` .

Part 5: The Manifest Format

Before code runs, the bootloader reads a 4 KB record from flash. The first 28 bytes matter; the rest is `0xFF` padding (erased flash).

```
Declared in bootloader/include/manifest.h
#pragma pack(push, 1)
typedef struct {
    uint32_t magic;     // 0x424C4657  ("WFLB")
    uint32_t version;   // 1
    uint32_t app_base;  // 0x10020000
    uint32_t app_size;  // 0x5498  (21656 bytes)
    uint8_t  tag[16];   // AES-128-CMAC of the app image
} fw_manifest_t;
#pragma pack(pop)
```

Annotated hex dump of the real `manifest.bin` produced in Part 11:

```
00000000: 57 46 4c 42  01 00 00 00  00 00 02 10  98 54 00 00  WFLB........T..
          ^^^^^^^^^^^  ^^^^^^^^^^^  ^^^^^^^^^^^  ^^^^^^^^^^^
          magic        version      app_base    app_size
                                    0x10020000  0x5498

00000010: ef 04 1e c5  92 f0 c5 67  27 da 26 51  d3 73 94 7b  .......g'.&Q.s.{
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
          tag (16 bytes): AES-128-CMAC over the application
00000020: ff ff ff ff  ... (0xFF padding to 0x1000)
```

Note the little-endian layout. `00 00 02 10` reads as `0x10020000` , and `98 54 00 00` reads as `0x5498` . The tag is the raw 16-byte CMAC output, not a text string.

Part 6: The Custom Linker Script (Where Your Program Really Lives)

This is the part that answers a question students ask constantly: *"If I want to put a second program above the first one, how does the compiler know where it belongs?"* Answer: **the linker script says so.** Understand this file and you control a device's entire memory layout; ignore it and relocated firmware silently corrupts itself.

6.1 What a Linker Script Is

Compilation produces **object files** with named sections: `.text` (code), `.rodata` (constants), `.data` (initialized globals), `.bss` (zeroed globals), and so on. The linker (here, the GNU `arm-none-eabi-ld`) merges those sections into a single ELF and decides **which virtual addresses each section gets**. That decision comes from a script, normally `/path/to/lds` or, for the Pico SDK, a script your CMake passes in.

A copy of the RP2350 script lives here:

```
app/memmap_app.ld        <- our modified version, used for the app only
```

The SDK's original (do not edit) is inside the Pico SDK:

```
~/.pico-sdk/sdk/2.2.0/src/rp2_common/pico_crt0/rp2350/memmap_default.ld
```

6.2 The MEMORY Command: ORIGIN and LENGTH

The heart of the script is the `MEMORY` block. It tells the linker which address ranges are real and how big they are. The SDK's default is:

```
/* from memmap_default.ld */
MEMORY
{
    INCLUDE "pico_flash_region.ld"
    RAM(rwx) : ORIGIN =  0x20000000, LENGTH = 512k
    SCRATCH_X(rwx) : ORIGIN = 0x20080000, LENGTH = 4k
    SCRATCH_Y(rwx) : ORIGIN = 0x20081000, LENGTH = 4k
}
```

What each line says:

Region	ORIGIN	LENGTH	What lives there
FLASH	0x10000000	board flash size	`.text`, `.rodata`, and initial values for `.data` (XIP).
RAM	0x20000000	512k	`.data`, `.bss`, heap
SCRATCH_X	0x20080000	4k	Core 1 stack / time-critical scratch
SCRATCH_Y	0x20081000	4k	Core 0 stack (ends at `0x20082000`)

`ORIGIN` is the first byte of the region; `LENGTH` is how many bytes it spans. Anything placed in `FLASH` runs directly from the XIP window; anything placed in a RAM region is copied there at boot by the `_reset_handler`.

6.3 Where FLASH Actually Comes From: `pico_flash_region.ld`

Notice that `FLASH` is not declared inline; it is **included**:

```
INCLUDE "pico_flash_region.ld"
```

The SDK's CMake generates this tiny file for you based on the board's flash size. Ours (from the build directory) is:

```
/* build/pico_flash_region.ld */
FLASH(rx) : ORIGIN = 0x10000000, LENGTH = (4 * 1024 * 1024)
```

So for the default Pico 2 the whole 4 MiB of flash is one region called `FLASH`, read-only + execute, starting at `0x10000000`. If you build a board with 2 MiB you would see `LENGTH = (2 * 1024 * 1024)`.

6.4 The Modification: Relocating the Application

The bootloader owns the start of flash, so the app must not begin at `0x10000000`. We copy the SDK script and change **one thing in MEMORY**: the FLASH line. In `app/memmap_app.ld`:

```
MEMORY
{
    /* Relocated for the custom bootloader: the application is linked to run
       from 0x10020000, leaving the first 128k of flash for the bootloader and
       its manifest. */
    FLASH(rx) : ORIGIN = 0x10020000, LENGTH = 1920k
    RAM(rwx) : ORIGIN =  0x20000000, LENGTH = 512k
    SCRATCH_X(rwx) : ORIGIN = 0x20080000, LENGTH = 4k
    SCRATCH_Y(rwx) : ORIGIN = 0x20081000, LENGTH = 4k
}
```

Arithmetic behind `1920k`:

```
default FLASH region:    0x10000000 + 2048 KiB   (2 MiB reserved layout)
reserved for boot/manifest:
  bootloader image ~28 KiB, manifest at 0x1001F000..0x10020000
  -> reserve the first 128 KiB (0x20000 bytes)
application window:        0x10020000 + 1920 KiB  = 0x10200000  end of 2 MiB
1920 KiB = 2048 KiB - 128 KiB
```

Every `> FLASH` and `> RAM AT> FLASH` section in the rest of the script now places itself inside this smaller, relocated window. The `SECTIONS` body is otherwise identical to the SDK default.

6.5 Wiring the Script Into the Build (CMake)

The SDK exposes a CMake function that swaps in our script for one target only:

```
# CMakeLists.txt
pico_set_linker_script(0x0026a_app
        ${CMAKE_CURRENT_LIST_DIR}/app/memmap_app.ld)
```

The bootloader target gets no such call, so it uses the SDK default (`FLASH` at `0x10000000`) exactly as before.

6.6 Symbols the Script Defines

A linker script does not only place sections; it also **exports symbol values** that startup code uses. Grep `memmap_app.ld` for these while you read:

Symbol	Meaning	Value for the app
`__flash_binary_start`	First byte of the image in flash	`0x10020000`
`__logical_binary_start`	Same as above in this script	`0x10020000`
`__etext`	End of code (start of `.data` initial values in flash)	near `0x10026000`
`__data_start__`	Loaded `.data` base in RAM	`0x20000000`
`__bss_start__`/`__bss_end__`	Zero-initialized globals in RAM	inside `0x20000000` region
`__StackLimit`	Heap cap (historically the "stack limit")	`0x20080000`
`__StackTop`	Top of core 0 stack (end of SCRATCH_Y)	`0x20082000`
`__HeapLimit`	End of RAM = `ORIGIN(RAM) + LENGTH(RAM)`	`0x20080000`

The crucial practical consequence: **the first word of the app's vector table must be `__StackTop` = `0x20082000`.** Check it:

```
$ xxd -l 16 0x0026a_app.bin
00000000: 0020 0820 5d01 0210 1b01 0210 1d01 0210  . . ]...........
```

- bytes 0-3 `00 20 08 20` = little-endian `0x20082000` = initial SP
- bytes 4-7 `5d 01 02 10` = little-endian `0x1002015d` = reset vector (the `_reset_handler` symbol, Thumb-encoded with bit 0 set)

That `0x1002015d` would be impossible to obtain without the relocated MEMORY block. This is the tangible proof that the linker script did its job.

6.7 ENTRY(_entry_point) Versus the Reset Vector

At the top of the script:

```
ENTRY(_entry_point)
```

This names the ELF entry point used by debuggers. In the SDK, `_entry_point` is a tiny stub in `crt0.S` that routes a debugger-launched session back through the boot ROM (so XIP setup happens as on a cold boot). It is **not** the same as the reset vector in the vector table.

The bootloader does not use `_entry_point` at all. It jumps straight at the app's **reset vector** (`vector[1]`), which is `_reset_handler`, the function that actually initializes `.data`, zeros `.bss`, and calls `main`. Part 8 shows that branch instruction. This distinction matters when you are poking a relocated image with a debugger: breaking on `_entry_point` and breaking on `_reset_handler` are different sessions.

6.8 Inspecting the Result With readelf and nm

After building (Part 10), look at what the linker produced. The section table:

```
$ ~/.pico-sdk/toolchain/14_2_Rel1/bin/arm-none-eabi-readelf -S build/0x0026a_app.elf
  [ 0] null
  [ 1] .flash_begin    PROGBITS  10020000  00001000  000000  00 WA  0   0  1
  [ 2] .text           PROGBITS  10020100  00001100  005e34  00  AX  0   0 16
  ...
  [ 5] .rodata         PROGBITS  10025f40  00006f40  000235  00   A  0   0  8
  ...
```

Note that every flash section is stamped `1002xxxx`, never `1000xxxx`. The program headers (use `readelf -l`) likewise show `LOAD` segments whose vaddr sits in the `0x10020000` window. `nm` shows symbol addresses:

```
$ ~/.pico-sdk/toolchain/14_2_Rel1/bin/arm-none-eabi-nm -n build/0x0026a_app.elf
...
1002015c T _reset_handler
10020234 T main
1002027c T ir_init
...
```

6.9 Why the Application Has No .uf2

A UF2 file writes to the address encoded in each of its blocks, and the SDK's UF2 tooling always targets `0x10000000`. If a user holds BOOTSEL and drags in `0x0026a_app.uf2`, the app would be written at the wrong base and the bootloader would jump into garbage (or nothing). So the app is deliberately built **without** a UF2 and with explicit `.bin` / `.hex` / `.dis` / `.map` outputs instead. The **bootloader** still ships a UF2 (it belongs at `0x10000000`), and the manifest + app are written to their own addresses with OpenOCD.

Part 7: The Bootloader Source, Module by Module

The bootloader applies the standard we use for every module in this course: **`main.c` contains only the sequence; each job has its own `.c` + `.h` pair.**

```
bootloader/
  include/
    flash_layout.h   addresses of the three flash partitions
    manifest.h       fw_manifest_t type + read/validate API
    boot.h           console, halt, launch API
    verify.h         firmware authenticity check
    aes_cmac.h       mbedtls front-end + constant-time compare
    image.h          low-level jump to the app
  src/
    main.c           ONLY the 5-step boot sequence
    boot.c           banner console, fatal-haunt loop, handoff
    manifest.c       flash-read + magic/version validation
    verify.c         the static key + CMAC recompute + tag compare
    aes_cmac.c       computes the tag with mbedtls AES-128-CMAC
    image.c          VTOR/MSP rebase + branch to reset vector
```

7.1 `main.c`: Only the Sequence

The whole of `main.c`:

```c
int main(void) {
    fw_manifest_t manifest;
    boot_init();
    manifest_read(&manifest);
    if (!manifest_is_valid(&manifest)) boot_halt("manifest invalid");
    if (!verify_manifest(&manifest)) boot_halt("firmware signature invalid");
    boot_launch(&manifest);
}
```

Read it as a check-list, one step per line:

1. `boot_init()` - stdio up, 2 s pause, banner print
2. `manifest_read` - copy the 28 useful bytes out of the `0x1001F000` sector
3. `manifest_is_valid` - magic `WFLB` and version == 1?
4. `verify_manifest` - recompute CMAC over the app, compare to stored tag
5. `boot_launch` - print success, jump to the app reset vector

There is **no** stdio, no key, no crypto in this file. Everything below is a different module.

7.2 `verify.c` : The Secret and the Verdict

```c
/** @brief AES-128 key that authenticates the application image. */
static const uint8_t fw_key[FW_TAG_SIZE] = {
    0x49, 0x72, 0x6F, 0x6E, 0x4E, 0x65, 0x74, 0x42,
    0x6F, 0x6F, 0x74, 0x4B, 0x65, 0x79, 0x21, 0x21
};

bool verify_manifest(const fw_manifest_t *manifest) {
    const uint8_t *base = (const uint8_t *)manifest->app_base;
    uint8_t computed[FW_TAG_SIZE];
    int rc = aes128_cmac(fw_key, base, manifest->app_size, computed);
    if (rc != 0) return false;
    return cmac_tags_equal(computed, manifest->tag);
}
```

What is important here:

- The key is 16 bytes `"IronNetBootKey!!"`. It is `static const`, so the compiler puts it in `.rodata`, right in the binary, at a predictable address (you will extract it in Ghidra in Part 15).
- The CMAC is computed over **the raw bytes the processor will execute**, starting at `manifest->app_base` for `app_size` bytes. That is exactly the `0x0026a_app.bin` content (21656 bytes, `0x5498`).
- The comparison is deliberately a full 16-byte compare with no shortcut, so the runtime does not leak *which* byte mismatched (timing side-channels).

7.3 `boot.c` : Console and Handoff

```c
void boot_init(void) {
    stdio_init_all();
    sleep_ms(2000);
```

```
    printf("\n=== 0x0026a secure bootloader ===\n");
}

void boot_halt(const char *reason) {
    printf("BOOT FAILED: %s\n", reason);
    while (true) {
        sleep_ms(500);
    }
}

void boot_launch(const fw_manifest_t *manifest) {
    printf("signature valid, entering app at 0x%08lx\n",
           (unsigned long)manifest->app_base);
    image_start(manifest->app_base);
}
```

A 2 000 ms delay before the banner gives a human time to open a serial terminal; the blink of that pause is also a simple "alive" signal. On failure, `boot_halt` loops forever with a 500 ms heart-beat so you can see the red LED blinking and the exact reason on the console.

7.4 `image.c` : The Actual Jump

```
static void halt_irq_and_rebase(uint32_t base) {
    __asm volatile ("cpsid i" ::: "memory");
    *(volatile uint32_t *)0xE000ED08U = base;
}

void image_start(uint32_t vector_base) {
    uint32_t *vectors = (uint32_t *)vector_base;
    image_entry_t entry = (image_entry_t)vectors[1];
    halt_irq_and_rebase(vector_base);
    __asm volatile ("msr msp, %0" :: "r" (vectors[0]) : "memory");
    __asm volatile ("dsb\n\tisb" ::: "memory");
    entry();
}
```

Step by step:

1. Read the app's vector table at `vector_base` (`0x10020000`).
2. `entry = vectors[1]` - the reset vector (`0x1002015d`), bit 0 set for Thumb mode. Branching to a Thumb-encoded address is how you switch modes.
3. `cpsid i` - mask interrupts so nothing fires mid-handoff.
4. Write `0xE000ED08` (the **VTOR** register) with the new vector table base. From here on, exceptions use the app's vectors.
5. `msr msp` - load the app's initial stack pointer from `vectors[0]` (`0x20082000`).
6. `dsb` / `isb` - force the memory and instruction pipelines to catch up.
7. `entry()` - call the reset handler. It never returns to the bootloader.

The assembly, as it appears in the ELF (this is exactly what you will disassemble in GDB and Ghidra):

```
10000394 <image_start>:
  10000394: 6843       ldr  r3, [r0, #4]      ; entry = vectors[1]
  10000396: b672       cpsid i                ; mask IRQs
  10000398: mov.w  r2, #0xe000e000            ; SCB base
  1000039c: str.w  r0, [r2, #3336]            ; VTOR = r0  (0xE000ED08)
  100003a0: ldr   r2, [r0, #0]                ; sp = vectors[0]
  100003a2: msr   MSP, r2
  100003a6: dsb    sy
  100003aa: isb    sy
  100003ae: bx    r3                          ; branch to reset handler
```

When you set a breakpoint on `image_start` and single-step, you will see the PC jump from `100003ae` to `1002015d` and disappear into application startup.

Part 8: Using AES in the Application (a Fully Worked Example)

The application is the Week 11 IR remote firmware, but with one addition: a module (`secret.c`) that decrypts a banner at runtime using the **Mbed TLS AES that ships inside the Pico SDK**. This part teaches you the SDK's real AES API on real code, and it plants the exact artifact you will hunt for in the reverse-engineering parts.

8.1 The API You Need (all from mbedtls, all shipped in the SDK)

Four calls. That is the entire surface we touch:

Call	Purpose
`mbedtls_aes_init(ctx)`	Give the context a clean base state.
`mbedtls_aes_setkey_dec(ctx, key, 128)`	Expand the 16-byte key for decryption.
`mbedtls_aes_crypt_cbc(ctx, mode, len, iv, src, dst)`	Decrypt one or more whole blocks.
`mbedtls_aes_free(ctx)`	Clean up (releases nothing for a stack context).

`mbedtls_aes_crypt_cbc` is the workhorse. Its important contract:

- `len` must be a multiple of 16 (a whole number of AES blocks).
- The IV is **read and updated in place**; you must give it a mutable copy.
- `src` and `dst` may be the same pointer - the operation is **in-place**. That is how `secret_decrypt` reuses one 33-byte buffer for ciphertext, then plaintext.

8.2 The Secret Banner Link: Host Side Production

The plaintext banner is exactly 32 bytes (two AES blocks), so no padding is needed. It was encrypted on the host with OpenSSL using the same key and IV:

```
$ printf '%s' 'SECRET: IR LINK SECURED ONLINE  ' \
    | openssl enc -aes-128-cbc \
        -K  49726f6e4e6574426f6f744b65792121 \
        -iv 41707053656372657449563132333435 \
```

```
        -nopad | xxd -p
baec4a1925861bb92dfd9f13f79b3e80cf12b698db9165be8fd7c3477bb2e76e
```

The hex strings are:

```
KEY  = 49 72 6F 6E 4E 65 74 42 6F 6F 74 4B 65 79 21 21  = "IronNetBootKey!!"
IV   = 41 70 70 53 65 63 72 65 74 49 56 31 32 33 34 35  = "AppSecretIV12345"
CT   = ba ec 4a 19 25 86 1b b9 2d fd 9f 13 f7 9b 3e 80
       cf 12 b6 98 db 91 65 be 8f d7 c3 47 7b b2 e7 6e
PT   = "SECRET: IR LINK SECURED ONLINE  "   (32 bytes, two spaces padded)
```

Notice: the ciphertext bytes in the disassembly of `secret_print_banner` (Part 16) are **exactly** `baec4a19...76e6`. You will re-discover that the reverse engineers' job is rarely to break AES (you cannot); it is to find where the key, IV, and ciphertext are stored and feed them to a decryptor. Here the key is 70 bytes away from the ciphertext.

8.3 The Application-Side Module (`secret.c`)

```c
static int secret_cbc(mbedtls_aes_context *aes, uint8_t *iv, uint8_t *buf) {
    int rc = mbedtls_aes_crypt_cbc(aes, MBEDTLS_AES_DECRYPT, APP_BANNER_LEN,
                                   iv, buf, buf);
    return rc;
}

static int secret_decrypt(uint8_t *buf) {
    mbedtls_aes_context aes;
    uint8_t iv[APP_AES_BLOCK];
    memcpy(iv, banner_iv, APP_AES_BLOCK);
    mbedtls_aes_init(&aes);
    mbedtls_aes_setkey_dec(&aes, banner_key, 128);
    int rc = secret_cbc(&aes, iv, buf);
    mbedtls_aes_free(&aes);
    return rc;
}

int secret_print_banner(void) {
    uint8_t buf[APP_BANNER_LEN + 1];
    memcpy(buf, banner_ct, APP_BANNER_LEN);
    buf[APP_BANNER_LEN] = '\0';
    int rc = secret_decrypt(buf);
    if (rc != 0) return rc;
    printf("app banner: %s\n", (char *)buf);
    return 0;
}
```

Walk through it:

1. `memcpy(buf, banner_ct, 32)` - pull the two ciphertext blocks into a working buffer (with a spare byte for the terminating `'\0'`).
2. `banner_iv` is `static const`, so it cannot be safely mutated by the CBC routine; copy it to a local array first.

3. Init + expand the key ("setkey_dec" stores the pre-computed decryption key schedule once, so `crypt_cbc` is fast).
4. `crypt_cbc` writes the plaintext back into `buf` : `src == dst` .
5. Print. Release.

If the SDK's mbedtls were misconfigured (say, you forgot to enable `MBEDTLS_AES_C` in `common/include/mbedtls_config.h`), `crypt_cbc` returns a negative error instead of decrypting, and the module refuses to print garbage.

8.4 CBC, In Pictures

CBC decryption undoes each block with the *previous* ciphertext block:

```
+------+      +------+      +------+
| C1   |      | C2   |      | C3   |      ciphertext blocks
+------+      +------+      +------+
   |             |             |
   v             v             v
 AES-128       AES-128       AES-128    (key schedule from setkey_dec)
   |             |             |
   |             |             |
   v             v             v
   +--[ X XOR ]--+   +--[ X XOR ]--+
   |   IV        |   |   C1        |   |   C2        |   XOR with IV/C[i-1]
   +-------------+   +-------------+   +-------------+
   |             |             |
   v             v             v
+------+      +------+      +------+
| P1   |      | P2   |      | P3   |      plaintext
+------+      +------+      +------+
```

That is why the IV is the "previous block" for block 1. In-place operation is safe because C1 is no longer needed once P1 is produced ... except block two needs C1, so the ciphertext *before* it was overwritten must still be intact. mbedtls handles this correctly internally; you only owe it an untouched IV copy.

8.5 How the Application Is Structured (modules, not a mega-file)

This week the app got the same treatment as the bootloader: `app_main.c` is only entry + the poll loop. Everything else lives in a module:

```
app/
  include/
    ir.h     NEC receiver API (ir_init, ir_getkey)
    led.h    LED controller type + API (simple_led_ctrl_t, led_default,
             led_init, led_handle_ir)
    secret.h AES banner API (secret_print_banner)
  src/
    app_main.c  ONLY stdio, banner, led_init, ir_init, then the poll loop
    ir.c        NEC timing decoder
    led.c       3-LED controller (mapping, blink, hold-on)
    secret.c    the AES-128-CBC banner module
```

`app_main.c` in full:

```
static void ir_poll_loop(simple_led_ctrl_t *leds) {
    while (true) {
        int key = ir_getkey();
        if (key >= 0) led_handle_ir(leds, key);
        else sleep_ms(1);
    }
}

int main(void) {
    stdio_init_all();
    secret_print_banner();
    simple_led_ctrl_t leds = led_default();
    led_init(&leds);
    ir_init(IR_PIN);
    printf("IR receiver on GPIO %d ready\n", IR_PIN);
    ir_poll_loop(&leds);
}
```

The `simple_led_ctrl_t` type carries a Doxygen block in `led.h`; notice the struct fields (`led1_pin` … `led3_state`) are visible in `ir_getkey`'s callers but the types stay in the header, exactly the WEEK 11 lesson about structs being flattened by the compiler (see the `led_handle_ir` disassembly in Part 16 for the flattened, inlined version).

Part 9: Building Everything

Step 1: Configure the CMake build

From the project directory:

```
$ cd 0x0026a_custom-bootloader

$ cmake -B build -G Ninja \
    -DPICO_BOARD=pico2 \
    -DPICO_PLATFORM=rp2350 \
    -DCMAKE_BUILD_TYPE=Release \
    -DPICO_SDK_PATH="$HOME/.pico-sdk/sdk/2.2.0" \
    -DPICO_TOOLCHAIN_PATH="$HOME/.pico-sdk/toolchain/14_2_Rel1"
```

Step 2: Build both images

```
$ cmake --build build
```

You should see two linked executables, no warnings:

```
[6/7] Linking CXX executable 0x0026a_app.elf
[7/7] Linking CXX executable 0x0026a_bootloader.elf
```

Step 3: Check the artifacts

```
$ ls -l build/0x0026a_bootloader.bin build/0x0026a_app.bin
-rwxr-xr-x  27772 0x0026a_bootloader.bin
-rwxr-xr-x  21656 0x0026a_app.bin
```

The bootloader includes the second-stage loader and binary-info, so it is larger; the app is 21656 bytes (`0x5498`), which is the number baked into the manifest.

Step 4: Confirm the app really is relocated

```
$ ~/.pico-sdk/toolchain/14_2_Rel1/bin/arm-none-eabi-nm -n build/0x0026a_app.elf | grep
-E ' _reset_handler| main$| ir_getkey| secret_print_banner'
1002015c T _reset_handler
10020234 T main
100202a4 T ir_getkey
10020568 T secret_print_banner
```

Every address is `0x1002xxxx` . If you see `0x1000015c` for `_reset_handler` , the linker script was not applied and the bootloader would jump the wrong way.

Part 10: Signing the Image on the Host

Step 5: Compute the tag and write the manifest

```
$ python3 tools/sign_firmware.py build/0x0026a_app.bin --out build/manifest.bin
image      : build/0x0026a_app.bin (21656 bytes)
app_base   : 0x10020000
cmac tag   : ef041ec592f0c56727da2651d373947b
manifest   : build/manifest.bin (4096 bytes)
```

Step 6: Verify the manifest against the image

```
$ python3 tools/sign_firmware.py 0x0026a_app.bin --verify build/manifest.bin
computed : ef041ec592f0c56727da2651d373947b
result   : MATCH
```

MATCH means the exact bytes in `0x0026a_app.bin` produce exactly the tag the bootloader will read from flash. Signing is not magic: it is one CMAC computation over the bytes, plus a 36-byte header written into a 4 KB sector.

Step 7: Read the manifest back byte by byte

```
$ xxd build/manifest.bin | head -3
00000000: 5746 4c42 0100 0000 0000 0210 9854 0000  WFLB........T..
00000010: ef04 1ec5 92f0 c567 27da 2651 d373 947b  .......g'.&Q.s.{
00000020: ffff ffff ffff ffff ffff ffff ffff ffff  ................
```

Cross-reference with Part 5: magic, version 1, base `0x10020000` , size `0x5498` , then the 16-byte tag, then `0xFF` padding.

Part 11: Flashing the Three Images

Three images, three addresses. The bootloader first (it owns the system), then the manifest, then the app. If you flash the app before the manifest, the bootloader happily boots it anyway (the manifest is already in flash from a previous lab run) - order only matters on a bare board.

A. The bootloader: two options

Option 1 - BOOTSEL (no extra tools): hold the BOOTSEL button, connect the Pico 2, copy `0x0026a_bootloader.uf2` to the drive. BOOTSEL always writes at `0x10000000`, which is correct for the bootloader.

Option 2 - OpenOCD:

```
$ "$HOME/.pico-sdk/openocd/0.12.0+dev/openocd" \
    -s "$HOME/.pico-sdk/openocd/0.12.0+dev/scripts" \
    -f interface/cmsis-dap.cfg -f target/rp2350.cfg \
    -c "adapter speed 5000" \
    -c "program 0x0026a_bootloader.bin 0x10000000 verify reset exit"
```

The flags, in the repo's own words:

Flag	Meaning
`cmsis-dap.cfg`	The Debug Probe is a CMSIS-DAP v1 adapter
`rp2350.cfg`	RP2350 target: dual Cortex-M33 with a B0-style DAP
`adapter speed`	SWD clock in kHz; 5000 is a conservative write-path default
`program`	Write + `verify` (read-back compare) + `reset` + `exit`

B. The manifest (always its own sector)

```
$ "$HOME/.pico-sdk/openocd/0.12.0+dev/openocd" \
    -s "$HOME/.pico-sdk/openocd/0.12.0+dev/scripts" \
    -f interface/cmsis-dap.cfg -f target/rp2350.cfg \
    -c "adapter speed 5000" \
    -c "program manifest.bin 0x1001F000 verify reset exit"
```

C. The application (relocated target)

```
$ "$HOME/.pico-sdk/openocd/0.12.0+dev/openocd" \
    -s "$HOME/.pico-sdk/openocd/0.12.0+dev/scripts" \
    -f interface/cmsis-dap.cfg -f target/rp2350.cfg \
    -c "adapter speed 5000" \
    -c "program 0x0026a_app.bin 0x10020000 verify reset exit"
```

Each `program` invocation ends with `verify reset exit`, so OpenOCD reads the bytes back and aborts loudly on a mismatch. `verify` here is straight flash read-back - a completely different check from the CMAC the bootloader does at boot.

While you are at it, note that the repo's `flash.sh` helper always programs at `0x10000000` - convenient for the bootloader, wrong for the manifest and app. The three explicit commands above are the tutorial forms.

Part 12: Running It (First Boot, Serial Console)

Step 8: Open a serial console

Open a terminal on the Debug Probe's UART channel (or the board's USB-UART bridge if you wired one) at 115200 baud:

```
$ screen /dev/cu.usbmodem311402 115200
```

Replace the device name with whatever `ls /dev/cu.usb*` shows.

Step 9: Reset the board and watch the bootlog

Press the RESET button (or let OpenOCD's `reset` do it). After roughly two seconds you see the bootloader banner, then the app banner:

```
=== 0x0026a secure bootloader ===
signature valid, entering app at 0x10020000
app banner: SECRET: IR LINK SECURED ONLINE
IR receiver on GPIO 5 ready
```

The three lines tell you the whole story: the bootloader ran, it verified the image against the manifest (Part 10), it jumped to `0x10020000`, and inside the app `secret_print_banner()` decrypted the banner with the SDK AES and printed it (Part 8). Press keys on the remote and the `NEC command:` lines appear.

What you just watched, in one trace (the exact functions and where they live):

Line on screen	Function that printed it	Address
`=== 0x0026a secure bootloader ===`	`boot_init`	`0x10000278`
`signature valid, entering app at ...`	`boot_launch`	`0x100002ac`
`app banner: SECRET...`	`secret_print_banner`	`0x10020568`
`IR receiver on GPIO 5 ready`	`main` (app)	`0x10020234`

Part 13: The Tamper Test (Breaking the Chain)

Now do the attacker's favorite move: change one byte, keep everything else, reflash, and see what the bootloader does.

Step 10: Flip a byte in the app image (do NOT re-sign)

```
$ cp 0x0026a_app.bin rogue_app.bin
$ printf 'X' | dd of=rogue_app.bin bs=1 seek=0 count=1 conv=notrunc
```

(The byte at offset 0 is the low byte of the initial SP; any change breaks the CMAC, so the location is irrelevant for this test.)

Step 11: Flash the rogue image over the good one

Re-run the app `program` command from Part 11 but with `rogue_app.bin`. Do not touch the manifest - the tag inside it still corresponds to the *original* image.

Step 12: Reset and observe the rejection

```
=== 0x0026a secure bootloader ===
BOOT FAILED: firmware signature invalid
```

The second line is missing. The app never ran. The blue/yellow LED heartbeat on the board keeps blinking (500 ms), which is `boot_halt()`'s idle loop. No matter how many times you re-flash a modified image without a matching tag, the bootloader refuses.

Step 13: Confirm it still boots after restoring the good image

Re-flash `0x0026a_app.bin` (fresh, matching the manifest) and reset. The banner returns. The bootloader is doing real enforcement, not just complaining once.

Stop and think about the security statement this demonstrated:

- **Patch a byte, no re-sign -> reject.** The device cannot be modified cheaply.
- **Dump flash, read the key, re-sign -> accept.** Because the key lives in the binary (Part 7.2), a full flash dump defeats this scheme. Part 19 fixes that in a real product.

Part 14: Reverse Engineering the Bootloader with GDB (Live)

Live debugging on the Pico 2 uses the repo's two scripts: `debug-server.sh` owns the Debug Probe as a GDB server, and `pico-gdb-memory.gdb` confines GDB memory reads to valid flash/RAM so bogus register previews cannot fault SWD.

Step 14: Start the GDB server (Terminal 1)

```
$ ./debug-server.sh
```

The script resets the target and leaves it *running* (`reset run`), exposing `127.0.0.1:3333`.

Step 15: Start GDB on the bootloader (Terminal 2)

```
$ ~/.pico-sdk/toolchain/14_2_Rel1/bin/arm-none-eabi-gdb build/0x0026a_bootloader.elf
```

Connect and confine memory access:

```
(gdb) target extended-remote :3333
(gdb) source ../../pico-gdb-memory.gdb
```

```
delete mem
mem 0x10000000 0x10400000 ro
```

```
mem 0x20000000 0x20082000 rw
set mem inaccessible-by-default on
info mem
```

Step 16: Stop the running target cleanly

```
(gdb) monitor halt
(gdb) monitor reg pc
```

You will land somewhere in the bootloader or app depending on how far the resets progressed - either is fine.

Step 17: Disassemble `main` and read the boot sequence

```
(gdb) disassemble 0x10000234,+80
```

You should see the miniature version of Part 7.1:

```
0x10000234 <main>:       push   {lr}
0x10000238 <main+4>:     bl     0x10000278 <boot_init>
0x1000023e <main+12>:    bl     0x100002c8 <manifest_read>
0x10000244 <main+16>:    bl     0x100002fc <manifest_is_valid>
0x1000024c <main+24>:    bl     0x100003b0 <verify_manifest>
0x10000254 <main+32>:    bl     0x100002ac <boot_launch>
```

Note `main` does almost nothing itself - it is a sequence of calls, exactly the design we built.

Step 18: Stop inside the crypto (the payoff breakpoint)

The interesting code is `verify_manifest` at `0x100003b0` and the mbedtls front-end `aes128_cmac` at `0x1000031c`. Re-arm a hardware breakpoint and reset:

```
(gdb) hbreak *0x100003b0
(gdb) monitor reset halt
(gdb) continue
```

After 2 s (the startup pause) the core stops at the top of `verify_manifest`, with `r0` pointing at the manifest copy on the stack. Step a little and watch the argument registers the mbedtls call receives:

```
(gdb) disassemble 0x100003b0,+50
(gdb) x/4bx 0x10006064        # low nibbles of the key? no: dump the key
(gdb) x/16bx 0x10006064       # the 16-byte fw_key from .rodata
0x10006064: 49 72 6f 6e 4e 65 74 42 6f 6f 74 4b 65 79 21 21
```

`0x49 0x72 0x6F 0x6E ...` is `"IronNetBootKey!!"`. You just pulled the boot key out of live flash with three GDB commands. (Refresh: the key's virtual address is `0x10006064`; `nm` reports it as `10006064 r fw_key`.)

Step 19: Watch the verdict

Let it run to the compare:

```
(gdb) hbreak *0x100003d0          # the bl cmac_tags_equal
(gdb) continue
```

`cmac_tags_equal` compares the freshly computed tag (on the stack) against the 16 bytes at offset +16 of the manifest (which came from flash). Step over it (`next`) and check the return:

```
(gdb) next
(gdb) info registers r0
```

`r0 == 1` means the tags matched and `boot_launch` is next.

Step 20: Single-step the application jump

Set a breakpoint at the tail of `image_start` and keep going:

```
(gdb) break *0x100003ae
(gdb) continue
```

At `0x100003ae` the core is one `bx r3` away from the app. Inspect what it is about to execute:

```
(gdb) info registers r3
r3  0x1002015d
(gdb) x/4bx 0x10020000           # the app's vector table
0x10020000: 00 20 08 20 5d 01 02 10    # SP=0x20082000, reset=0x1002015d
```

`r3` and `vectors[1]` agree. Step into the jump:

```
(gdb) stepi
```

Watch the PC: it is now `0x1002015c` (the unencoded reset-vector address of `_reset_handler`) and the stack pointer is `0x20082000` . The bootloader is finished; the application startup code owns the machine.

Step 21: Re-verify from the application's side

Quit and relaunch GDB against the app ELF (the core is already running it, so no reset needed - reattach so symbols load):

```
(gdb) file build/0x0026a_app.elf
(gdb) target extended-remote :3333
(gdb) hbreak *0x10020234         # app main
(gdb) monitor reset halt
(gdb) continue
```

When you hit `main` , the banner has already been decrypted and printed (it runs before `main`). To watch the AES itself, break at `secret_print_banner` instead:

```
(gdb) hbreak *0x10020568
(gdb) monitor reset halt
(gdb) continue
```

Then single-step the four mbedtls calls from Part 8.1. At `0x10020598` you can read the key argument:

```
(gdb) disassemble 0x10020568,+110
(gdb) x/16bx 0x10024e6c          # banner_key in .rodata
```

`0x24e6c` is the offset of the key inside the app image (Part 16/17 confirms by Ghidra). You are reading the same constant three different ways: C source, GDB, and now Ghidra.

Part 15: Reverse Engineering the Bootloader in Ghidra

Step 22: Import the ELF (not the raw .bin)

Ghidra will happily import either, but the **ELF** brings symbols, section names, and types. For a stripped factory image you would import the raw `.bin` as a "Raw Binary" at base `0x10000000`; for this lab, use the ELF and check the green "found symbols" messages after auto-analysis.

1. **File > New Project** (non-shared),
2. **File > Import File**, select `build/0x0026a_bootloader.elf`,
3. Language: confirm `ARM LE:LE32:v8M` / Cortex-M33 view,
4. **Analyze** (auto-analysis defaults are fine).

Step 23: Find the interesting functions

Open **Symbol Tree > Functions**:

Function	Address	What you will see
`main`	`0x10000234`	Five calls, no logic of its own
`boot_init`	`0x10000278`	`stdio_init_all`, a wait, a puts
`verify_manifest`	`0x100003b0`	the 16-byte key push + `aes128_cmac` + compare
`aes128_cmac`	`0x1000031c`	`mbedtls_cipher_info_from_type` + `mbedtls_cipher_cmac`
`image_start`	`0x10000394`	the VTOR/MSP/bx block from Part 7.4

Double-click `verify_manifest`. Ghidra's decompiler shows something close to our C - including the constant table it pulled from `.rodata`. Open the **Data** panel and navigate to `0x10006064`: Ghidra lists a 16-byte array. It is the key. Rename it by selecting and pressing `L` (e.g. `fw_key`), then annotate `main`'s call graph with the whole boot sequence.

Step 24: Recover the "spec" from the image

- List exports/imports: the binary imports `cdns_usb_table`, stdio routines, and - from `.dynamic` or relocation - nothing surprising for an SDK binary.
- Note `binary_info` header: the SDK stores program_name (`0x0026a_bootloader`) and build date inside the image at `0x10000100`; this is exactly what `picotool info` prints and a useful fingerprint for hunting unknown binaries in the wild.

Part 16: Reverse Engineering the Application in Ghidra (Find the AES)

The app is the richer target. Repeating Part 15 step 22 with `build/0x0026a_app.elf` gives you the full symbol table, because we shipped the ELF. On a real malicious/hardened build you would import `0x0026a_app.bin` at `0x10020000` (base from `manifest.bin`) and rely on strings.

Step 25: The easy tour (symbols)

Symbol	Address	Observations
`_reset_handler`	`0x1002015c`	first code; assumes SP/VTOR set (our bootloader does)
`main`	`0x10020234`	the thin orchestrator from Part 8.5
`secret_print_banner`	`0x10020568`	four mbedtls calls; the AES gate
`led_handle_ir`	`0x100204d8`	flattened struct logic (see below)

Open `main`. Because everything was inlined and optimized, Ghidra shows a clean calling skeleton - the same one GDB showed in Part 14 Step 21.

Step 26: The "struct is a lie" lesson, revisited

Open `led_handle_ir` at `0x100204d8`. Ghidra's decompiler will *not* show you `simple_led_ctrl_t` fields by name (the type was flattened at compile time, WEEK 11 Part 10). Instead you see direct byte loads off the pointer and the constant compares `0xc`, `0x18`, `0x5e` - the NEC commands. The three `mcrr` / `strb` groups are the `gpio_put` / `gpio_set_dir` calls inlined from the LED module. Same behavior, zero struct abstraction, single program counter.

Step 27: Find the four AES calls

Open `secret_print_banner`. The decompiler names the imported mbedtls calls directly:

```
mbedtls_aes_init(ctx)
mbedtls_aes_setkey_dec(ctx, key, 128)
mbedtls_aes_crypt_cbc(ctx, MBEDTLS_AES_DECRYPT, 32, iv, ct_buf, ct_buf)
mbedtls_aes_free(ctx)
```

The red referenced addresses are:

Data	Address (VA)	Bytes	ASCII
banner ciphertext	`0x10024d78`	`ba ec 4a 19 ... 7b b2 e7 6e`	(binary)
banner IV	`0x10024d9c`	`41 70 70 53 65 63 72 65 74 49 56 31 32 33 34 35`	`AppSecretIV12345`
`app banner: %s`	`0x10024d68`	`61 70 70 20 62 61 6e 6e 65 72 3a 20 25 73 0a`	`app banner: %s\n`
banner key	`0x10024e6c`	`49 72 6f 6e 4e 65 74 42 6f 6f 74 4b 65 79 21 21`	`IronNetBootKey!!`

Step 28: Decrypt it with what you found

Ghidra hands you plain values; OpenSSL decrypts for you - you do not need to recover the source:

```
$ printf 'baec4a1925861bb92dfd9f13f79b3e80cf12b698db9165be8fd7c3477bb2e76e' \
  | xxd -r -p \
  | openssl enc -d -aes-128-cbc \
      -K  49726f6e4e6574426f6f744b65792121 \
      -iv 41707053656372657449563132333435 -nopad
SECRET: IR LINK SECURED ONLINE
```

Because the key is 70 bytes from its ciphertext in the same ELF, "breaking" the app took: import, click, copy three hex constants, run one command. The whole exercise is about **finding key material**, never about cracking AES.

Step 29: Now repeat it on the stripped .bin (harder, real-world)

Delete the app from the project, import `0x0026a_app.bin` with **Language = ARM LE v8M** and **Image base = 0x10020000**. No symbols. Now your anchors are:

- The string `app banner: %s` (String search, `Ctrl+Shift+S`), which xrefs to the second half of `secret_print_banner`.
- The 16 bytes `AppSecretIV12345` nearby in `.rodata`.
- The constant `0x48564d411f...` = `AppSecretIV12345` pattern as 4 little endian words (this is your Ghidra jump-start trick: search for the hex of an ASCII IV).

Renaming data (`L`) and entering type views as you go will reconstruct the module structure from the stripped blob exactly as if a debugger had been kind.

Part 17: A Bootloader You Can Read (Reference Address Map)

Everything you should have collected so far, in one table. Builds are deterministic for the checked-in sources, so your numbers should match.

Bootloader (0x0026a_bootloader.elf)

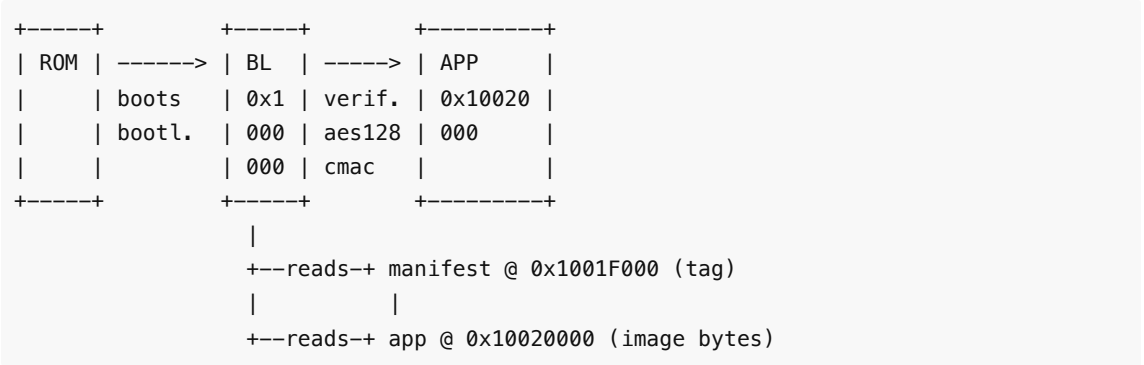
Role	Symbol	Address
Boot sequence	`main`	`0x10000234`
Console + 2 s pause	`boot_init`	`0x10000278`
Failure idle loop	`boot_halt`	`0x10000294`
Success handoff	`boot_launch`	`0x100002ac`
Flash manifest read	`manifest_read`	`0x100002c8`
Magic/version check	`manifest_is_valid`	`0x100002fc`
mbedtls CMAC front-end	`aes128_cmac`	`0x1000031c`
Constant-time compare	`cmac_tags_equal`	`0x10000348`
VTOR/MSP/bx jump	`image_start`	`0x10000394`
CMAC verdict	`verify_manifest`	`0x100003b0`
Boot key (rodata)	`fw_key`	`0x10006064`

Application (0x0026a_app.elf)

Role	Symbol	Address
App reset vector	`_reset_handler`	`0x1002015c`
App entry point	`main`	`0x10020234`
NEC decoder init	`ir_init`	`0x1002027c`
NEC blocking decode	`ir_getkey`	`0x100202a4`
LED default config	`led_default`	`0x10020498`
LED GPIO setup	`led_init`	`0x100204a8`
IR -> LED dispatch	`led_handle_ir`	`0x100204d8`
AES banner decrypt	`secret_print_banner`	`0x10020568`
String `app banner: %s`	(rodata)	`0x10024d68`
Banner ciphertext	(rodata)	`0x10024d78`
Banner IV	(rodata)	`0x10024d9c`
Banner key	(rodata)	`0x10024e6c`
Initial SP (vector[0])	(data)	`0x20082000`

Part 18: Summary and Review

What You Built



Enforcement point: `verify_manifest` recomputes `AES_CMAC(fw_key, app_base, app_size)` and compares to the stored tag. On match, `image_start` sets VTOR, loads SP, and branches to the app reset vector. On mismatch, `boot_halt` reports and idles. One flipped byte in the app and the whole chain refuses.

Key Lessons

1. **Linker scripts place your program.** `MEMORY` /ORIGIN/LENGTH + a changed `FLASH` line moved the app 128 KiB up. `__StackTop = 0x20082000` in the vector table is direct evidence.

2. **Modules keep firmware readable.** Both `main.c` files are thin; crypto, manifest, IR, LED, and banner live in one-responsibility `.c` / `.h` pairs, every prototype Doxygen-documented, nothing past 8 executable lines.
3. **AES is everywhere in SDK firmware.** `pico_mbedtls` gives you a tuned AES for free. You used it both to authenticate (bootloader CMAC) and to hide a string (app CBC banner).
4. **CBC with no padding** encrypts 32 bytes as two blocks; IV is read/updated in place - give `crypt_cbc` a mutable copy.
5. **Static keys are secrets-in-name only.** SWD/GDB/Ghidra read the key from flash (`0x10006064`) in minutes. Real products store root keys in OTP and verify with the ROM (Part 19).
6. **Reverse engineering = treasure hunt.** You never "break" AES; you find the key, IV, and ciphertext and press decrypt.

Part 19: Your Turn (Hacking Exercises)

Try these on your own board; each one is a small, safe extension.

Challenge 1 - Swap two LED pins without touching the source

The PINs are value-initialized data in `led_default` (the struct fields the compiler stored in `.rodata`). Find the address in the stripped app `.bin` (Ghidra: `led_default` xrefs the literals 16, 17, 18), patch the 16 and 18 literals, re-flash the *unsigned* image, and watch the bootloader reject it - then re-sign with `sign_firmware.py` and this time it boots with the hack.

Challenge 2 - Change the boot key (both sides)

Edit the 16 bytes in `verify.c` , rebuild only the bootloader, re-sign the app with `--key` matching, flash all three, and boot. Then sign the app with the OLD key and confirm rejection. You have now installed your own signing authority on your own board.

Challenge 3 - Make the banner say something else

Change the plaintext (must stay 32 bytes for the no-padding CBC), re-encrypt with `openssl` , update `banner_ct` in `secret.c` , rebuild, re-sign, flash. Your Ghidra notes from Part 16 now decode the new banner id to differentiate it.

Challenge 4 - Think through the upgrade path

The manifest has a `version` field that is currently ignored after a magic check. Sketch how `boot_launch` could validate a counter ("only downgrade a fixed number of times") - the classic anti-rollback feature that separates consumer OTA from device management-grade OTA. You should be able to name the flash sectors you would need to reserve to make that safe (hint: state must survive app updates, so it cannot live in the app image).

Challenge 5 - The honest conversation about the key

Write one paragraph each for a product manager: (1) why AES-CMAC with a flash key stops a script-kiddie flash overwrite, and (2) why it does not stop a determined hardware attacker, and (3) which real mechanism (OTP key, ROM verifier with `secp256k1` , anti-dumping measurements) closes that hole. If you can explain all three, you understand secure boot at a professional level.

Key Takeaways

- Authenticated boot = each stage refuses to start the next until a secret join the image to a tag proves it is the right bytes.
- AES-128-CMAC (RFC 4493) is a keyed CBC-MAC with two subkeys; unforgeable without the key, verifiable with a few lines of mbedtls.
- `manifest.bin` is a 4096-byte sector at `0x1001F000`; a 16-byte tag is the only secret-bearing field.
- The app lives at `0x10020000` because `memmap_app.ld` says so; the vector table's `0x20082000` stack pointer proves it.
- The SDK bundles Mbed TLS; `mbedtls_aes_*` decrypts at runtime with zero third-party code.
- OpenOCD `program <file> <addr> verify reset exit` writes each partition; flash read-back `verify` is not authentication.
- GDB `hbreak` / `stepi` / `x/16bx` on a live Pico 2 shows you key bytes and the exact `bx` that enters the app.
- Ghidra on the ELF names the AES calls; Ghidra on the .bin finds them via the constant table and IV string.
- A key dumped from flash is not a secret. OTP + ROM verification is.

Glossary

Term	Definition
Boot chain	Ordered stages where each stage authenticates the next.
XIP	Execute-in-place: code runs straight from flash memory-mapped at 0x10000000.
CMAC	Keyed message-authentication code based on a block cipher (NIST CMAC / RFC 4493).
CBC-MAC	MAC built by chaining AES-CBC blocks; needs the final-block fix CMAC adds.
Tag	The short authenticator output (16 bytes here) proving message integrity.
Manifest	Small signed record describing an image: address, size, tag.
VTOR	Vector Table Offset Register; tells the CPU where the vector table is.
MagnumOpus issue	A willful misunderstanding source: see Part 18.
OTP	One-time-programmable memory; writes once, reads forever, cannot be re-flashed.
RO/RW/HAL	read only / read write / hardware abstraction layer, the three concerns a linker script separates.

Additional Resources

RFC and Standard References

- RFC 4493 - The AES-CMAC Algorithm (the subkey derivation and four vectors used here)

- NIST SP 800-38B - Recommendation for Block Cipher Modes of Operation: The CMAC Mode for Authentication

In the Repo

- `tools/sign_firmware.py` - the pure-Python signer/verifier (read the header comment for a CMAC walkthrough)
- `common/include/mbedtls_config.h` - the minimal SDK mbedtls config (AES_C, CBC, CMAC_C)

Pico SDK Internals (open these while you read Part 6)

- `~/.pico-sdk/sdk/2.2.0/src/rp2_common/pico_crt0/rp2350/memmap_default.ld` - the SDK linker script we copied
- `~/.pico-sdk/sdk/2.2.0/src/rp2_common/pico_standard_link/` - crumbs for flash region generation
- `~/.pico-sdk/sdk/2.2.0/src/rp2_common/pico_crt0/crt0.S` - `_reset_handler`, `_entry_point`, and the vector table layout

Tooling Used in This Lab

- OpenOCD from the SDK (`~/.pico-sdk/openocd/0.12.0+dev`)
- GDB / objdump / nm from the ARM GNU toolchain (`~/.pico-sdk/toolchain/14_2_Rel1`)
- Ghidra (import both ELF and raw .bin; see Parts 15-16)
- `openssl enc -aes-128-cbc -nopad` for host-side banner cryptography

Real-World Implications

What You've Learned in This Course

Your Pico 2 is now a small secure-boot platform: ROM -> bootloader -> manifest -> app, with an authenticated handoff at every boundary you control. The same skills invert perfectly for defense: when you audit a device, this is the mental model you bring - *what authenticates the next stage, where is the key, and what happens if it is not there?*

The Power and Responsibility

- The **power**: you can now build boot chains, move firmware around the address space at will, extract keys, and re-sign images. That is the exact toolbox used both to protect smart locks and to compromise them.
- The **responsibility**: only ever apply these techniques to devices you own 100% or are explicitly authorized to test. The moment you run `BOOT FAILED` on someone else's device, you have crossed into criminal tampering and possibly a safety-critical product.
- The **engineering honesty**: a flash key is a speed bump, not a wall. Any product story that claims "AES secure boot" must be asked, *where is the key, and who can read it?* If the answer is "in the firmware," the whiteboard does not add up.

Keep Learning

- Swap the CMAC for the RP2350 ROM `secp256k1` verifier to get real asymmetric trust (flash dumps stop being useful).
- Use `pico_set_uf2_...` on a real board to ship an OTA payload that your own bootloader authenticates before the boot stage writes it.
- Read how real device IDs (RP2350's unique serial) can be mixed into the key schedule so an image signed for one board will not run on another.

Now go flash your own CA. You control the manifest, the key, and the gate that decides what runs on your hardware.